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India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Session : Technical paper

The Sentient Grid: How Agentic AI Is Creating the Autonomous Utility of the Future

Presented By

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Agentic AI is Moving Fast—and Utilities are leading the charge

Global Market

$10.7B by 2034

Agentic AI in Energy & Utilities is projected to grow at a 36.4% CAGR—signaling massive disruption and opportunity.

$42.7B market value in AI Spending in energy distribution market

Driven directly by grid modernization and renewable integration.

North America Leads

Commanding 35.2%+ of the global Agentic AI tools market in 2024.

Indian Market

Rapid growth of digital grid infrastructure in Indian markets

Indian grid automation market reached ~$1.8B in 2025 and is expected to grow to $3B by 2034.

Utilities are moving from Pilot projects → scaled AI Adoption

Indian Smart Grid Forum identified 174AI/ML use cases for utilities including 45 already implemented.

Utilities are moving towards intelligent, self optimizing networks

Early adopters are already seeing strong returns from Agentic AI implementations.

Source:

https://market.us/report/agentic-ai-tools-market/
https://market.us/report/agentic-ai-in-energy-market/#overview
https://medium.com/%40mrinmoy.aicch/agentic-ai-for-the-utility-industry-a9063e39a1da

Power Industry Outlook

- **Rising electricity demand driven by AI data centers**
- **Renewable Energy Expansion with increased grid complexity**
- **Innovative financing and contracting methods**
- **Artificial Intelligent led Utility Transformation**
- **Large Scale government push for AI**

Where Agentic AI is gaining traction

Generation

- **Behind the meter** Distributed Energy Resources
- **New generation in rate base**, even in deregulated markets

Transmission & Distribution

- Design criteria and robust cybersecurity
- **Grid-Enhancing Technologies** (GETS)
- Efficiently enable **new business connections**

Customer Service

- Expanded **self-service options**
- Improved **interconnection** processes

Regulatory & Legislative Strategy

- Bring Deep Expertise to Look Ahead
- Engage with **broad groups** of stakeholders

Relevance

In the race towards a decentralized and decarbonized grid, Agentic AI acts as the essential catalyst for unlocking grid intelligence and modernizing operations

How Agentic AI Is Creating the Autonomous Utility of the Future

From grid to customer, Agentic AI is reimagining how utilities work, respond, and grow. Agents can be used across Operations, Asset management and customer experience.

By automating billing, detecting anomalies, and personalizing client interactions, we drive operational excellence

Redefine the customer journey with real-time messaging and proactive alerts. By providing always-on support

Navigate policy shifts with ease. By automating reporting and dynamically adapting to environmental laws

Optimize every field technician's day and build a more resilient, high-performing operation with intelligent predictive maintenance and automated scheduling

Predicts outages, coordinates teams, and automates recovery workflows Reduces downtime and improves response agility when every minute matters

Drivers for successful implementation of Agentic AI in utilities

Gartner forecasts high failure rates (40%) in agentic AI by 2027 – utilities must build the right foundation to scale successfully

Robust Data Foundation

Real-time access to structured and unstructured data across operational and customer systems

Business Value Alignment

AI use cases must directly link to cost savings, customer satisfaction, or reliability improvements.

Cross-Functional Buy-In

Alignment across cross functional departments like IT, operations, customer care, and regulatory teams to define objectives and integration paths.

Scalable Architecture

Modular, cloud-native platforms with APIs for seamless integration across utility systems (e.g., GIS, SCADA, MDMS)

Change Management & Talent Enablement

Accelerating workforce readiness and overcoming resistance with immersive training, iterative pilots, and transparent communication.

Human-in-the-Loop Design

Empower your workforce with AI agents designed to augment human expertise, governed by robust oversight to ensure absolute trust, transparency, and accountability.

Predictive Asset Management:
A "Maintenance Agent" continuously monitors thermal stress on distribution transformers.
The Action: Instead of just sending an alert, it autonomously raises a digital work order, checks spare part inventory, and assigns the nearest field crew via a mobile app.

Intelligent Revenue Assurance:
A "Billing Agent" analyses AMI (Advanced Metering Infrastructure) data to detect consumption anomalies.
The Action: It identifies potential "behind-the-meter" theft or meter tampering and triggers an automated remote disconnection or a physical inspection task for high-loss feeders.

Real-time Grid Balancing (VPPs):
A "Virtual Power Plant (VPP) Agent" aggregates thousands of small-scale rooftop solar and EV batteries across a city like Delhi or Bangalore.
The Action: It bids this "stored energy" into the ancillary services market during peak evening hours, creating a new revenue stream for the utility.

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)